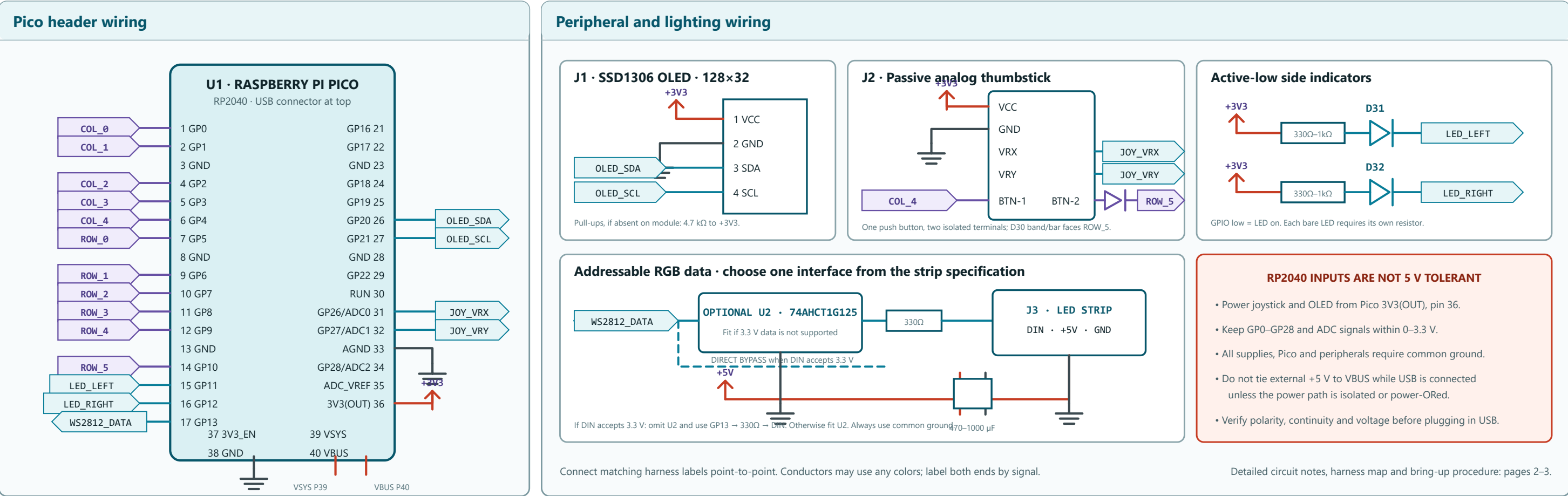
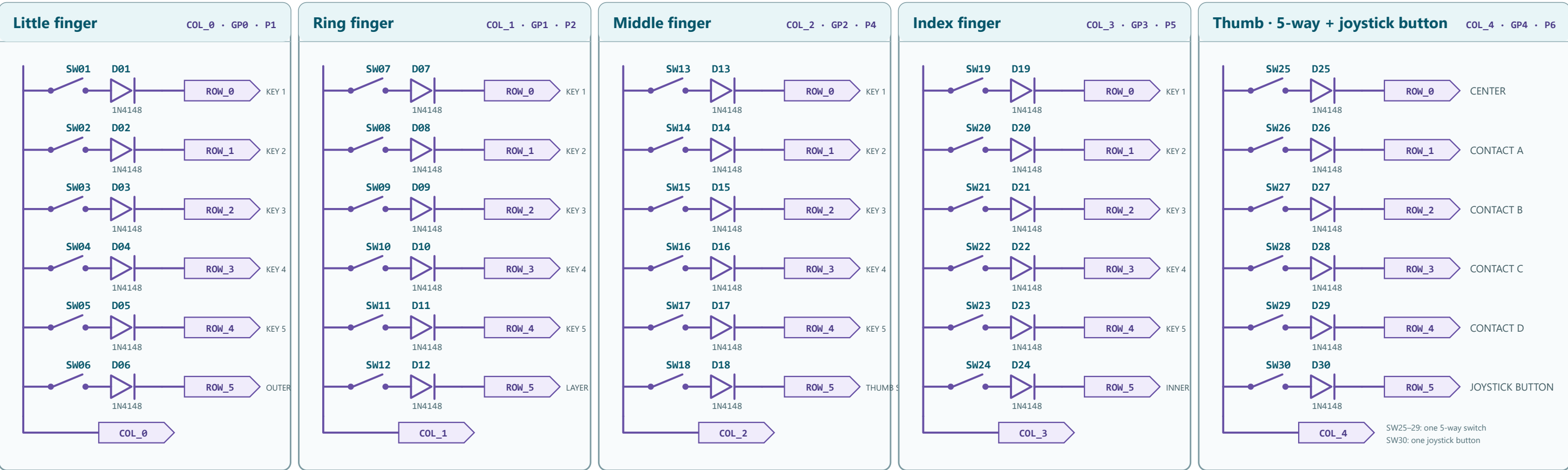


REPLICAZERON · RP2040 WIRING DIAGRAM

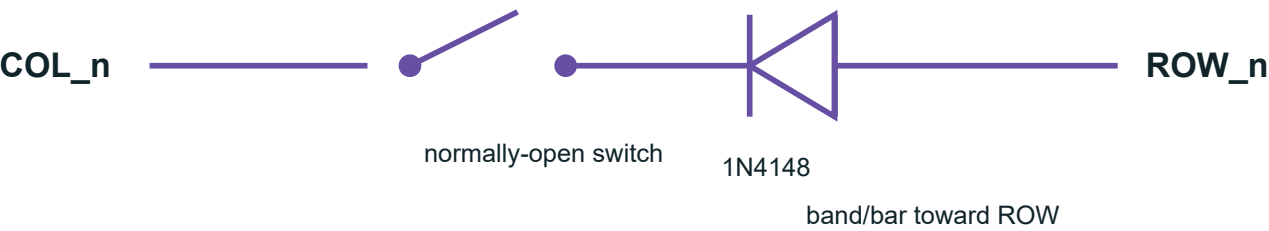
Point-to-point assembly wiring · standard Raspberry Pi Pico · sheet 1/1



Peripheral circuits and power

Recommended implementation details for a reliable, serviceable build

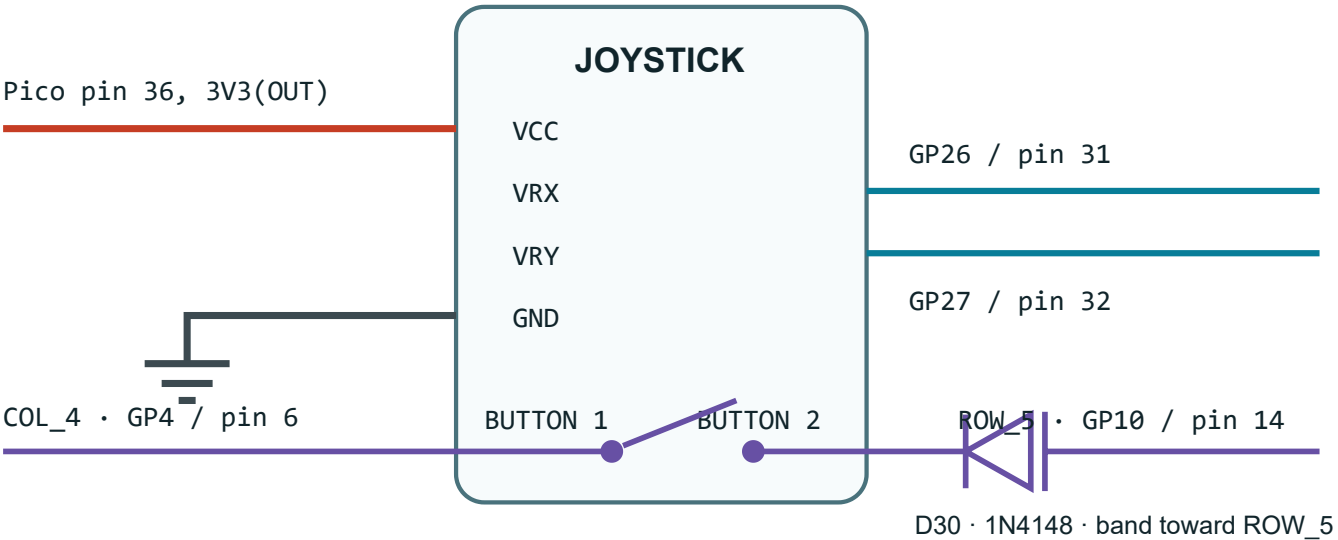
Matrix switch cell — repeat for every populated key



COL bus – switch – diode anode → diode cathode/bar – ROW bus

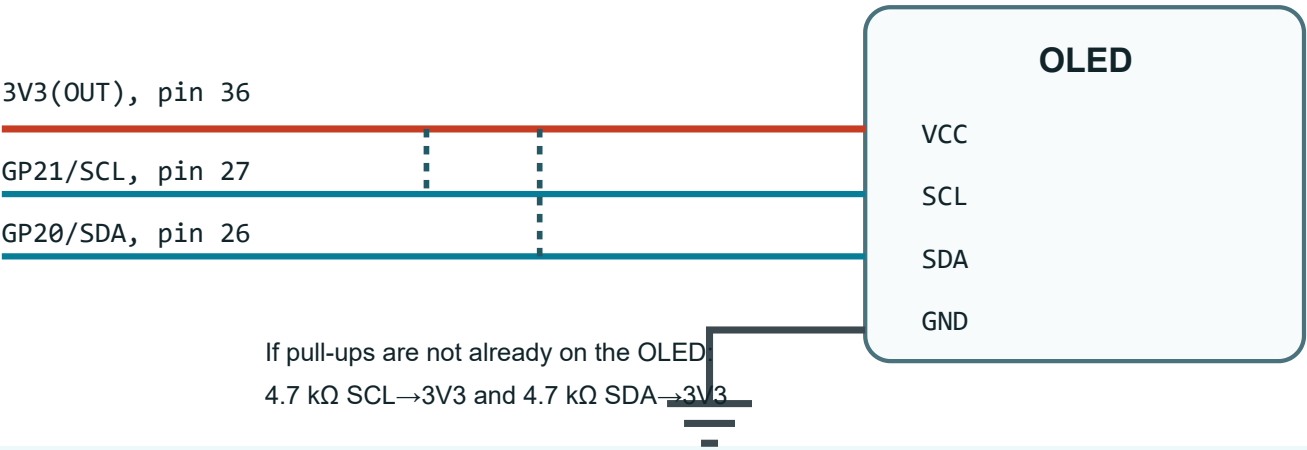
Do not share one diode between switches. All diodes must face the same way.

Analog joystick — 3.3 V only



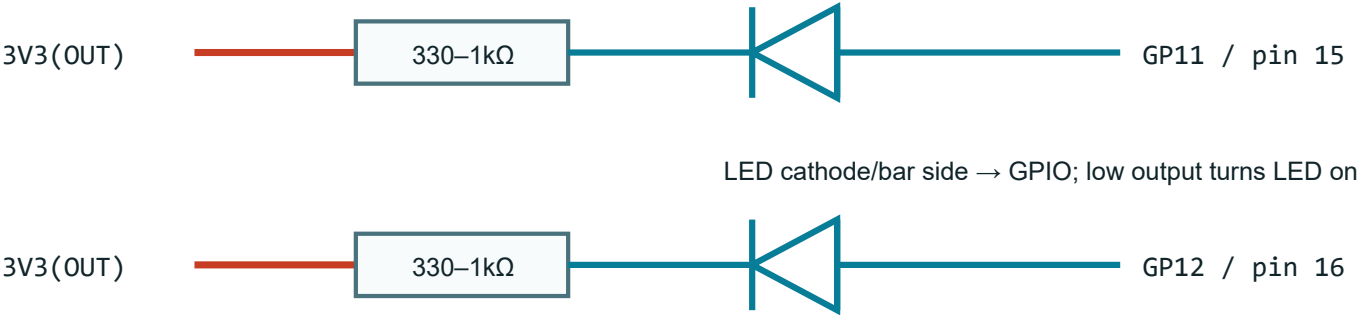
Never connect the joystick VCC wire to 5 V when VRX/VRY go directly to GP26/GP27. Measure both axis voltages first: they must stay between 0 and 3.3 V over full travel. There is one joystick push button with two terminals. Those terminals must be isolated; do not connect a module's GND-referenced SW output directly as a matrix key.

OLED I²C

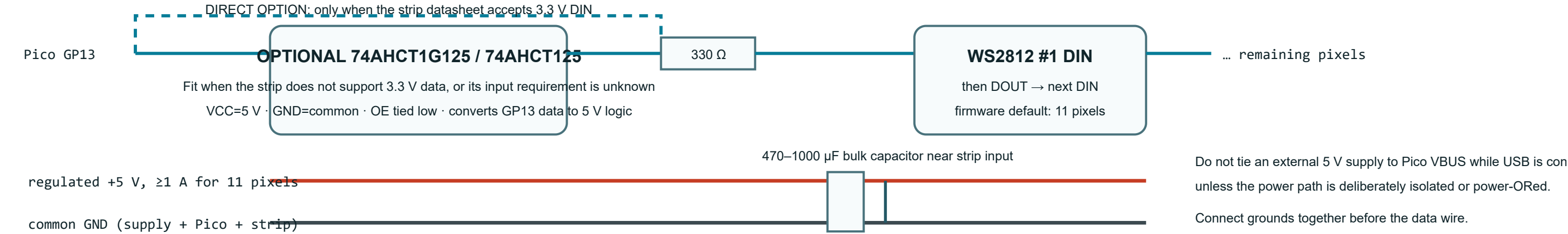


Power the module from 3.3 V unless its exact schematic proves that 5 V VCC cannot pull SDA/SCL above 3.3 V.

Side indicators — active low, matching 9R



WS2812 strip — choose the data interface from the strip specification



Harness map and bring-up checklist

Use continuity mode with all power disconnected. Choose any conductor colors, then label and verify every signal.

Matrix harness

Signal	Pico GPIO	Physical pin
COL_0	GP0	1
COL_1	GP1	2
COL_2	GP2	4
COL_3	GP3	5
COL_4	GP4	6
ROW_0	GP5	7
ROW_1	GP6	9
ROW_2	GP7	10
ROW_3	GP8	11
ROW_4	GP9	12
ROW_5	GP10	14

Peripherals

Function	Pico	Physical
Left side LED	GP11	15
Right side LED	GP12	16
WS2812 data	GP13	17
OLED SDA	GP20	26
OLED SCL	GP21	27
Joystick VRX	GP26/ADC0	31
Joystick VRY	GP27/ADC1	32
Joystick ground	AGND	33
Joystick click	COL_4 + ROW_5	GP4 + GP10
3.3 V peripherals	3V3(OUT)	36
Logic ground	GND	3/8/13/18/23/28/38

GP14, GP15, GP16–GP19, GP22, and GP28 remain spare in this firmware.

Reported 5-way wiring

Contact	Matrix connection
Common	COL_4 / GP4
Center	ROW_0 / GP5
A	ROW_1 / GP6
B	ROW_2 / GP7
C	ROW_3 / GP8
D	ROW_4 / GP9

Direction mapping to resolve: 9R assigns the five contacts to ROW_0–ROW_4 but does not label their physical directions. The supplied harness calls ROW_0 Center, while current OLED navigation calls (0,4) Right and (1,4) Select. Identify A–D with a meter, then align firmware or wiring.

Power-off checks

1. Disconnect USB and every external supply.
2. Check for a short between 3V3 and GND, then 5V and GND.
3. Verify BOOTSEL is not mechanically stuck. BOOTSEL is onboard and must not be wired into the key matrix.
4. Continuity-check every harness wire to the Pico physical pin—not only the GP label.
5. Verify every matrix diode band faces the ROW bus.
6. Verify GP0 and GP5 are not permanently shorted together.

Safe staged bring-up

1. Disconnect OLED, joystick, WS2812 strip, and side LEDs. Connect only Pico USB.
2. Flash the UF2 and confirm the RPI-RP2 drive disappears and the USB HID device enumerates.
3. Add the matrix harness. Test one switch at a time in Vial/WebHID.
4. Add joystick powered from 3V3. Measure VRX/VRX before connecting GP26/GP27.
5. Add OLED at 3.3 V and confirm SDA/SCL idle near 3.3 V.
6. Add side LEDs with resistors.
7. Add the externally powered WS2812 strip last. Connect GP13 directly through 330 Ω only if the strip supports 3.3 V data; otherwise fit the level shifter.

Design basis: compared net-by-net with 9R's STM32 KiCad schematic. It preserves the COL2ROW topology, COL_4 thumb common, ROW_5 stick click, 3.3 V analog/OLED supplies, active-low indicators, and 5 V WS2812 supply while translating MCU pins to Pico. The level shifter is optional when the exact strip supports a 3.3 V data input; otherwise it is required for dependable level conversion. The data resistor, bulk capacitor, and separated LED-power guidance are reliability additions. This is not a certification drawing; confirm actual module datasheets.